

Service Operations Management



UNIT – IV

Forecasting Demand for Services: A review of different types of forecasting methods for demand forecasting.

Managing Capacity and Demand: Strategies for matching capacity and demand, Psychology of waiting, Application of various tools used in managing waiting line in services.

Managing Facilitating Goods: Review of inventory models, Role of inventory in services

Managing service supply relationship: Understanding the supply chain/hub of service, Strategies for managing suppliers of service

Vehicle Routing Problem: Managing after sales service, Understanding services that involve transportation of people and vehicle, Techniques for optimizing vehicle routes

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UNIT – IV: Service Operations Management (Descriptive Long Answers)

1. Forecasting Demand for Services

Introduction

Demand forecasting is the process of estimating future customer demand for a service. Accurate forecasting helps service organizations allocate resources, schedule employees, manage inventory, and maintain customer satisfaction. Unlike manufacturing, service demand is often variable, perishable, and difficult to store, making forecasting extremely important.

Types of Forecasting Methods

A. Qualitative Forecasting Methods

These methods rely on human judgment, experience, and expert opinions.

1. Delphi Method

- Experts provide forecasts independently.
- Responses are collected and analyzed.

- Feedback is shared anonymously.
- Process continues until consensus is reached.

Advantages

- Useful when historical data is unavailable.
- Reduces personal bias.

Applications

- New service launches.
- Long-term planning.

2. Market Research Method

- Surveys, interviews, and focus groups are conducted.
- Customer preferences are analyzed.

Example

A hospital surveys patients before introducing telemedicine services.

3. Executive Opinion Method

- Forecasts are based on managers' collective experience.

Advantages

- Quick and inexpensive.

Disadvantages

- May be subjective.

B. Quantitative Forecasting Methods

These methods use historical data and mathematical models.

1. Moving Average Method

Forecast is calculated using the average of previous observations.

Formula

$$\text{Forecast} = \frac{\sum \text{Demand}}{\text{Number of Periods}}$$

Advantages

- Simple.
- Smooths random fluctuations.

2. Exponential Smoothing

Recent observations receive greater weight.

Formula

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

Where:

- F_t = Forecast
- A_t = Actual demand
- α = Smoothing constant

3. Trend Analysis

Uses historical patterns to predict future demand.

Example

Increasing airline passenger traffic over several years.

4. Regression Analysis

Establishes relationships between variables.

Example

Hotel bookings may depend on tourism season, weather, and holidays.

Importance of Demand Forecasting

- Better capacity planning
- Efficient manpower scheduling
- Improved customer service
- Reduced costs
- Better resource utilization

Conclusion

Demand forecasting enables service organizations to anticipate customer needs and allocate resources efficiently, resulting in higher service quality and profitability.

2. Managing Capacity and Demand

Introduction

Capacity refers to the maximum service output that can be produced during a given period. Demand refers to customers' need for the service. Since services cannot be stored, balancing capacity and demand becomes crucial.

Strategies for Matching Capacity and Demand

A. Managing Demand

1. Differential Pricing

Charging different prices at different times.

Example

- Airline ticket pricing
- Hotel seasonal pricing

2. Reservations and Appointments

Control customer arrival patterns.

Example

- Doctor appointments
- Restaurant reservations

3. Promotion During Off-Peak Periods

Encourages customers to use services during low-demand periods.

Example

Cinema discounts on weekdays.

4. Demand Shifting

Customers are encouraged to use services during less busy periods.

B. Managing Capacity

1. Flexible Workforce

Employ part-time workers during peak demand.

2. Cross-Training Employees

Employees perform multiple tasks.

3. Shared Capacity

Resources are shared among departments.

4. Technology Utilization

Use self-service kiosks and online systems.

Psychology of Waiting

Customer satisfaction depends not only on actual waiting time but also on perceived waiting time.

Principles of Waiting Psychology

1. Unoccupied Time Feels Longer

Customers feel waiting is longer when they are idle.

2. Uncertain Waits Feel Longer

Unknown waiting times increase frustration.

3. Unexplained Waits Feel Longer

Customers want reasons for delays.

4. Unfair Waits Feel Longer

Queue jumping causes dissatisfaction.

5. Anxiety Makes Waits Longer

Concern about service quality increases perceived waiting.

Waiting Line (Queuing) Tools

1. Single Queue System

One queue serves multiple service counters.

Example

Banks.

2. Multiple Queue System

Separate queue for each server.

Example

Supermarkets.

3. Priority Queue

Certain customers receive priority.

Example

Emergency patients in hospitals.

4. Self-Service Systems

Customers perform services themselves.

Example

ATM machines.

Benefits

- Reduced waiting time
- Increased efficiency
- Improved customer satisfaction

Conclusion

Balancing demand and capacity helps organizations provide efficient service while minimizing customer waiting and resource wastage.

3. Managing Facilitating Goods

Introduction

Facilitating goods are physical items that support service delivery. Examples include medicines in hospitals, food ingredients in restaurants, and spare parts in repair services.

Inventory management ensures the right goods are available at the right time.

Role of Inventory in Services

1. Ensures Service Continuity

Prevents service interruptions.

2. Improves Customer Satisfaction

Required materials are readily available.

3. Reduces Stock-Outs

Avoids loss of customers.

4. Supports Operational Efficiency

Smooth service operations.

Inventory Models

1. Economic Order Quantity (EOQ)

Determines optimal order quantity.

$$EOQ = \sqrt{\frac{2DS}{H}}$$

Where:

- D = Annual demand
- S = Ordering cost
- H = Holding cost

Benefits

- Minimizes inventory costs.
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2. Reorder Point (ROP)

Determines when a new order should be placed.

$$ROP = \text{Demand during Lead Time}$$

3. ABC Analysis

Inventory classified into:

- A Items: High value, low quantity
 - B Items: Medium value
 - C Items: Low value, high quantity
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4. Just-In-Time (JIT)

Materials arrive exactly when needed.

Advantages

- Reduced inventory holding costs.
 - Improved efficiency.
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Conclusion

Effective inventory management ensures uninterrupted service delivery while minimizing costs and improving customer satisfaction.

4. Managing Service Supply Relationship

Introduction

A service supply chain consists of organizations, people, information, and resources involved in delivering services to customers.

Unlike manufacturing, service supply chains focus heavily on information and customer interaction.

Understanding the Service Supply Chain

Components

1. Service Provider

Organization delivering the service.

2. Suppliers

Provide resources and support.

3. Customers

Receive and evaluate services.

4. Information Systems

Coordinate activities among participants.

Strategies for Managing Service Suppliers

1. Supplier Selection

Choose suppliers based on:

- Quality
- Cost
- Reliability

- Delivery performance

2. Strategic Partnerships

Develop long-term relationships.

3. Performance Monitoring

Evaluate suppliers using KPIs.

4. Information Sharing

Exchange demand and inventory information.

5. Risk Management

Maintain backup suppliers.

Benefits of Effective Supplier Management

- Better service quality
- Reduced costs
- Improved reliability
- Faster service delivery
- Competitive advantage

Conclusion

Strong supplier relationships enhance service quality and ensure smooth operation of service supply chains.

5. Vehicle Routing Problem (VRP)

Introduction

The Vehicle Routing Problem (VRP) is a logistics and transportation optimization problem that determines the most efficient routes for vehicles serving customers.

The objective is to minimize transportation cost, travel time, and fuel consumption while maximizing service quality.

Importance of VRP

- Reduces operating costs
 - Improves delivery performance
 - Enhances customer satisfaction
 - Optimizes resource utilization
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Managing After-Sales Service

After-sales service includes activities provided after product purchase.

Examples

- Product installation
- Maintenance
- Repairs
- Technical support
- Warranty services

Importance

- Customer retention
 - Brand loyalty
 - Increased customer satisfaction
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Transportation of People and Vehicles

Services Involving Transportation of People

Examples:

- Airlines
- Railways
- Bus services
- Ride-sharing services

Services Involving Transportation of Goods

Examples:

- Courier services
- Logistics companies
- Food delivery services

Techniques for Optimizing Vehicle Routes

1. Route Scheduling

Determine best sequence of customer visits.

2. GPS-Based Navigation

Provides real-time route guidance.

3. Geographic Information Systems (GIS)

Analyzes road networks and traffic conditions.

4. Vehicle Tracking Systems

Monitor vehicle movement.

5. Optimization Algorithms

- Nearest Neighbor Method
- Clarke-Wright Savings Algorithm
- Genetic Algorithms
- Ant Colony Optimization

6. Dynamic Routing

Routes adjusted based on real-time traffic.

Benefits of Route Optimization

- Lower fuel consumption
- Reduced travel time
- Improved vehicle utilization
- Better customer service
- Reduced environmental impact

Conclusion

Vehicle routing is a critical aspect of service operations involving transportation and after-sales support. Proper route optimization improves efficiency, reduces costs, and enhances customer satisfaction.

Overall Conclusion of Unit IV

Unit IV focuses on managing service operations efficiently through **demand forecasting, capacity planning, waiting-line management, inventory control, supplier relationship management, and vehicle routing optimization**. These concepts help service organizations improve operational efficiency, reduce costs, increase customer satisfaction, and gain a sustainable competitive advantage.